BSc Project Screening Form:

Part 1 – Project Proposal

Student Name	Senod Hansindu Weerathunga
Student Number	2433323
Degree Pathway (course)	BSc (Hons) Computer Science
Supervisor Name	Ms. Dilushinie Fernando
Title of Project	**ElderConnect:** Developing a Social Engagement and Wellness Platform Using Artificial Intelligence for Elderly People
Abstract of the project	The aim of this project is to develop a mobile application which is simple and smart design for elderly individuals aged 60 years and above, providing an all-in-one digital platform to connect with friends, view personalized news, and manage their health. Many elderly people experience loneliness, difficulties using modern platforms, and challenges in maintaining their regular health routines. This tailored system proposes a friendly and interactive environment that helps users to stay socially connected, mentally active, and health conscious. The application features a colour-coded interface with large icons and guidance for ease of use. This includes two main interfaces, one for elderly users and another for guardians or caretakers, which is selectable during the registration. Elderly users can select their interests such as in news, hobbies, or sports, which an Artificial Intelligence (AI) recommendation model use to personalize their feed. This application provides a social feed, a customized news section, and a text-to-speech function for visually impaired users. An AI based mood detection model analyzes user's posts and reactions to identify emotions such as happiness, sadness, or neutrality. With consent, this data appears on the caretaker dashboard to help monitor emotional well-being of the user. The app also includes a health and medication reminder system and a wellness section featuring simple memory games, breathing exercises, and relaxation activities. Existing platforms like Facebook, GrandPad, and Connect2Affect by AARP are either too complex, require special devices or focuses only on social interaction without addressing mental wellness or health management. This proposed AI-based platform fills these gaps by combining easy communication, mood detection, personalized content, wellness and health management in one simple, accessible mobile app designed specifically for elderly users.
Project deliverables	**Software Deliverables** A mobile application for elderly users that enhances social interaction, emotional well-being, and health management with using AI features. • Elderly Portal: A simple interface for connecting with friends, personalized news, and wellness activities. Including AI mood detection, text-to-speech for the need, and medication reminders. • Guardian Portal: Allow to monitor elder user's mood, activity, and medication, and manage health data such as dosage and schedules.

	Document Deliverables • Project Proposal • Monthly Progress Report • Contextual Report • Reflective Report • Thesis Report
Description of your artefact	**Aim** To develop an **AI-based social and wellness mobile platform** that promotes social interaction, emotional well-being, and health awareness among elderly users through a simple, accessible, and personalized interface.
	Objectives • To design and implement an intuitive and accessible mobile interface tailored for elderly users. • To develop an AI-based sentiment analysis module for mood detection through text. • To personalize news and content recommendations based on user interests. • To integrate a text-to-speech feature for visually impaired users. • To include a wellness and mindfulness section with relaxation and memory activities. • To implement a medication reminder system connected to a caretaker interface. • To develop a caretaker dashboard for tracking user mood, medication adherence, and daily activity.
	List of Features • Simplified, color-coded signup process with clear visual and voice guidance. • Interest-based content personalization for news, hobbies, and entertainment. • Social feed for sharing posts, photos, and greetings with friends. • AI mood detection analyzing user messages and posts to assess emotional state. • A Text-to-speech functionality for visually impaired users. • “Talk Button” feature for sending short voice messages easily. • Wellness section with memory games, breathing exercises, and relaxation activities. • Medication reminders using visual or voice notifications. • Caretaker dashboard to manage medication schedules and monitor mood or activity. • Emergency contact button for quick connection with family or caretaker.

Risk analysis

Risk	Likelihood	Impact	Mitigation
Difficulty in integrating AI sentiment model due to API limitations	Medium	High	Use pre-trained APIs like Google Cloud NLP or using rule-based sentiment analysis.
Elderly users find the interface confusing or hard to navigate	High	High	Conduct usability testing and refine UI/UX for simplicity
Data privacy issues or unauthorized data access	Medium	High	Implement Firebase Authentication, encrypted storage, and restricted data sharing
Internet dependency may affect notifications or updates	Low	Medium	Enable offline caching and local notifications
Guardian dashboard synchronization errors	Low	Medium	Use Firebase real-time database with retry logic for data sync

How does your project relate to your degree course and build upon the units/knowledge you have studied/acquired

This project directly relates to the modules of Human-Computer Interaction (HCI), Mobile Application Development, and Database Systems. It demonstrates the integration of AI and design principles which are accessible to solve a real-world problem through innovative software engineering practices.

The key intellectual and technical components include integrating AI-based sentiment/mood analysis, ensuring real time data synchronization between users and caretakers, and developing a simple yet intelligent interface tailored for the elderly users.

Resources required in developing the artefact

Yang, J., Yang, T. and Li, L. (2025) *Analysis of user behavior and satisfaction under the elderly adaptation mode of an app based on the fuzzy-IPA model*, *Nature News*. Available at: https://www.nature.com/articles/s41598-024-84526-6 (Accessed: 23 October 2025).

Additional information in part 2

Have you completed & submitted your ethics form?

YES	NO

If the project is a development of previous work by yourself or others, give details below. Failing to declare such previous work here may be treated as an academic offence

Recoverable Signature

X

Dilushinie Fernando

Supervisor Signature: Signed by: 1a69e3a7-04c8-461a-969e-db0152e0be28

After the proposal has been signed off by both the supervisor and course coordinator scan the proposal and upload on BREO with signatures. Projects that follow proposals that have not been approved may be cancelled and there will be no compensation for any time lost

Part 2 – List of relevant resources

Fill in this section after your project proposal has been approved by your supervisor. Use Harvard referencing (see https://lrweb.beds.ac.uk/a-guide-to-referencing). Modify the list below as appropriate. This list is part of Assignment 1 and will be submitted with the Project Proposal.

1. Books

Pears, R. and Shields, G. (2019). Cite them right: The essential referencing guide. 11th edn. London: Bloomsbury Academic.

Gaver, W. W. et al. (2013). Designing for the digital age: How to create human-centered products and services. New Jersey: Wiley.

2. Journal Papers

Chua, C.S., Lim, W.M. and Teh, P.-L. (2023) 'Designing age-friendly mobile apps: Insights from a Mobility App Study', *Activities, Adaptation & Aging*, 48(3), pp. 409–429. doi:10.1080/01924788.2023.2213028.

Kawaguchi, K. *et al.* (2024) 'Effects of a mobile app to promote social participation on older adults: Randomized controlled trial', *Journal of Medical Internet Research*, 26. doi:10.2196/64196.

3. Web Sites with relevant information

Sen, K., Prybutok, G. and Prybutok, V. (2021) *The use of digital technology for social wellbeing reduces social isolation in older adults: A systematic review*, *SSM - population health*. Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC8733322/ (Accessed: 25 October 2025).

Socialize anywhere: Apps for senior connections (no date) *Care 365 Homecare In New York - Homecare Always Available*. Available at: https://www.care365.care/resources/socialize-anywhere-apps-for-senior-connections (Accessed: 25 October 2025).

4. Relevant software

a. Firebase Realtime Database. Google. Available at: https://firebase.google.com/products/realtime-database (Accessed 29 Oct 2025).

b. Hugging Face Sentiment Analysis API. Available at: https://huggingface.co/models (Accessed 30 Oct 2025).

5. Relevant hardware

a. Smartphones with accessibility features for elderly users (e.g., large screens, voice commands).

b. Personal computers or laptops for system development and testing.

6. Other

a. Design guidelines for elderly-friendly user interfaces – internal project documentation and usability research reports.

b. *Connect2Tools to overcome social isolation* (2025) *Connect2Affect*. Available at: https://connect2affect.org/ (Accessed: 26 October 2025).